Data persistence in the event driven environment of parcel sorting

Speakers

Jean-Richard Meinrad
Solution Architect

Personal information: 40 years old, 2 kids, from Aarau

Hobbies: Snowboard touring, Gardening, Home automation tinkering

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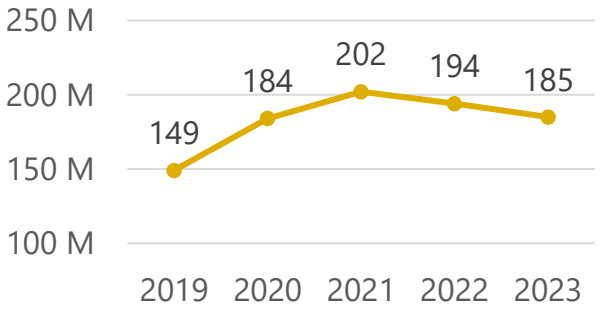
DevOps Engineer

Personal information: 32 years old, 1 dog, from Bern

Hobbies: Jogging, Gaming, Gym

Who we are working for



Logistic-Services 2023

185 M
Parcels delivered

1647 M
Letters posted and delivered in Switzerland

38,9 M
Imported mailpieces

131,8 M CHF
Invested in new parcel centers[1] and a new parcel sorting machine[2]

2040
Planned to be climate-neutral

[1] Pratteln (BL) and Wallisellen (ZH)
[2] In Parcelcenter Härkingen (SO)

Project and Team

WeSort (Weiterentwicklung Sortierung)
The current sorting solution is end-of-life.
Complete redesign of architecture for sorting mailpieces

- In-house development
- Centralized, modular and flexible
- Cost reduction
- Reduced workload
- Reduce time to market

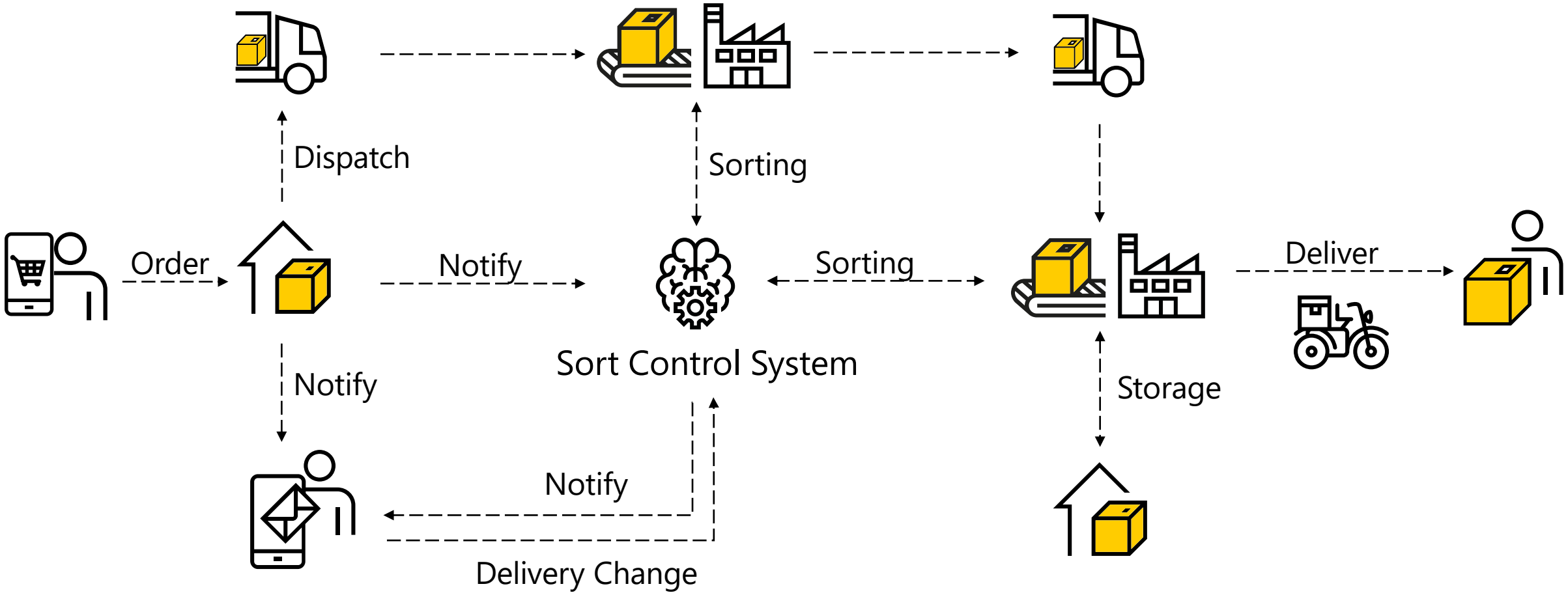
Sorting Software Solutions
Creating software solutions to prepare, plan and control mailpiece sorting in Switzerland.

15	Developers
4	Business Analysts
3	Solutions Architects
2	Project Owner
1	Scrum Master

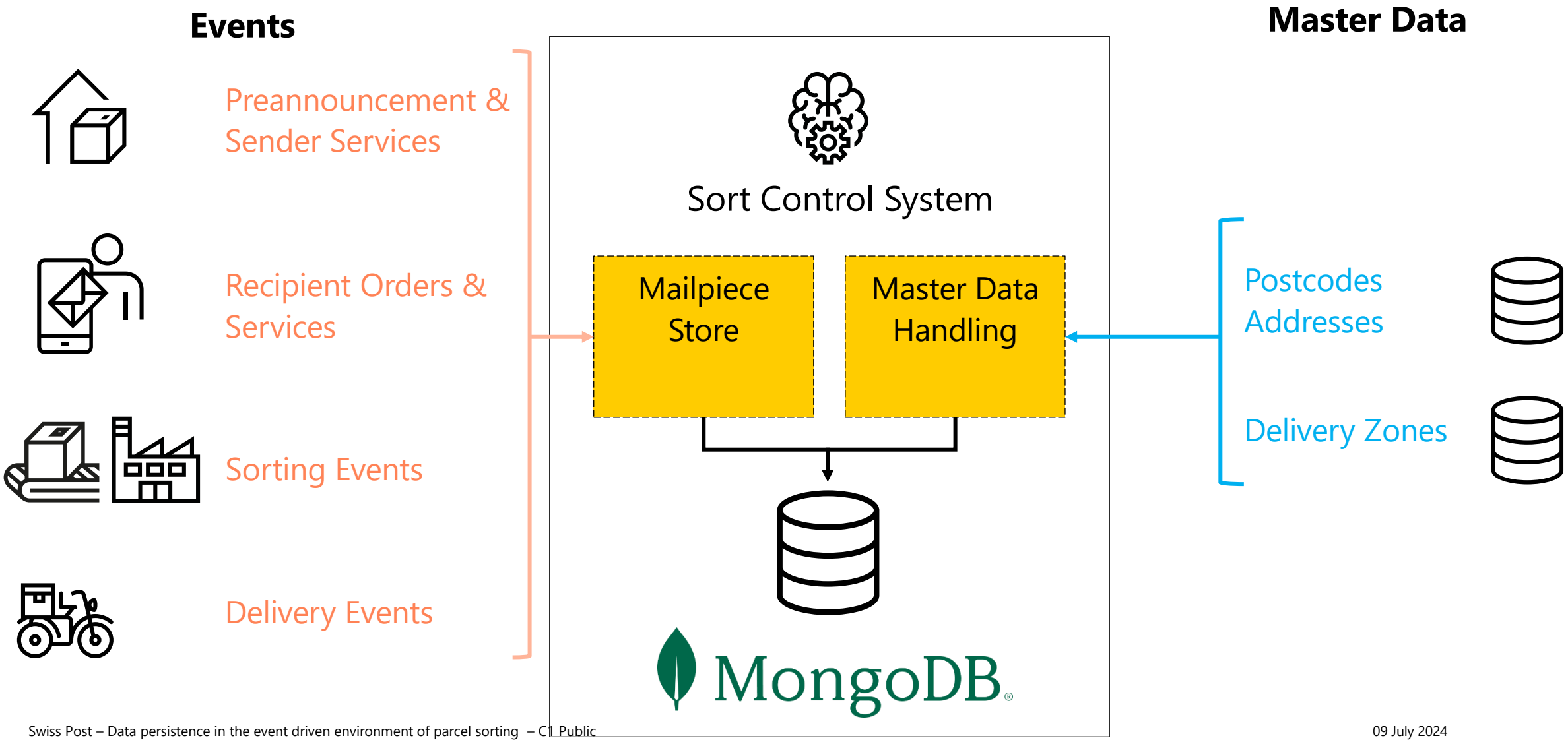
DevOps, 24/7 Support, CI/CD, Automation, Cloud, High-Quality, Monitoring & Traceability

How does parcel delivery work?

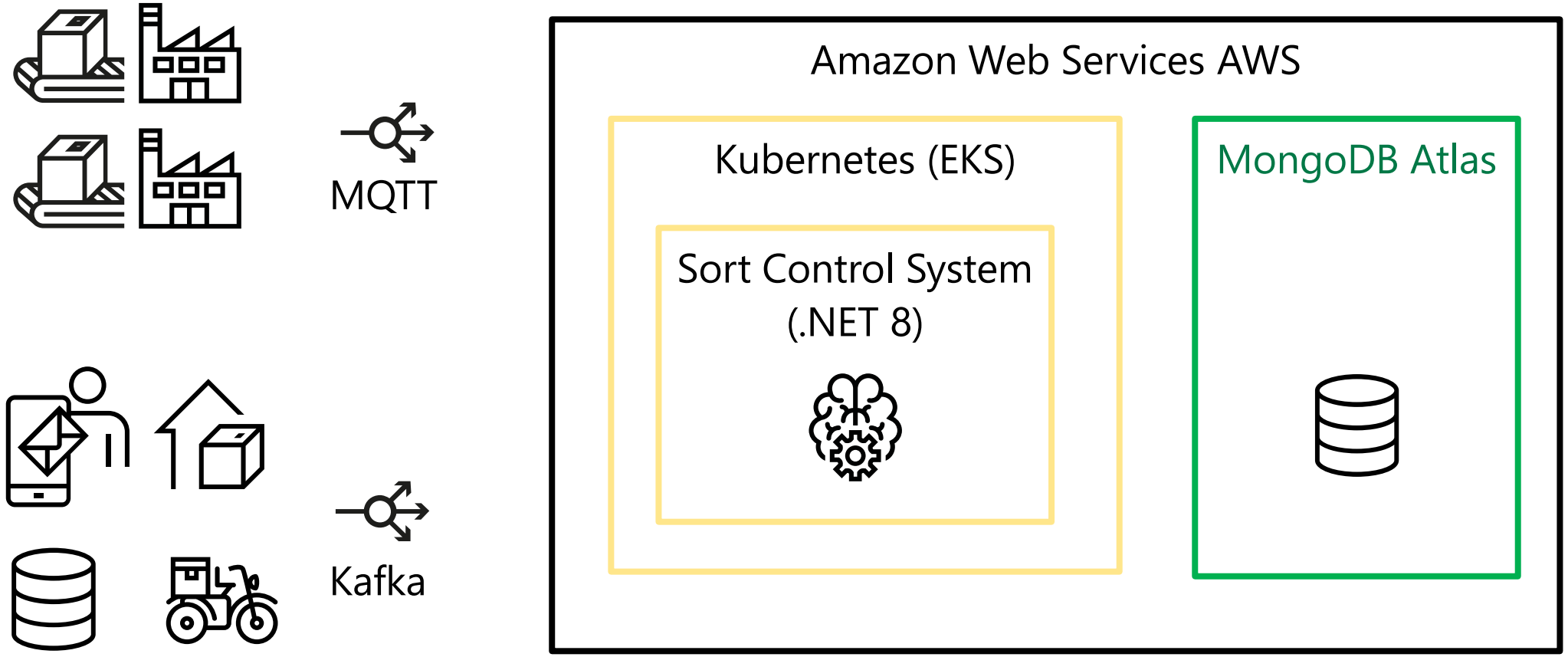
Use case: Order parcel online



Use case: Data and Events



Sorting System Overview



Mailpiece Store

Mailpiece Store

Goals:
- We want all information for a mailpiece in one place
- One lookup to retrieve all data

Challenges:
- Multiple producers of events
- Evolving contracts (breaking changes over time, invalid data, etc.)

Solution:
- Leverage MongoDB as a schemaless database
- Aggregate information in one document per mailpiece
- No multi document transaction necessary (push into array)

Mailpiece Store

```
{
  "MailpieceIdentifier": "9805031823036847 63",
  "UpdatedDate": "2024-05-14",
  "Events": [
    {
      "EventType": "Preannouncement",
      "SchemaValidationResult": {
        "IsValid": false,
        "ValidationErrors": [...],
        "ValidatedVersion": "0.3.3.0"
      },
      ...
    },
    {
      "EventType": "SortedEvent",
      ...
    },
    ...
  ]
}
```

- One Document per mailpiece
- `MailpieceIdentifier` for identification
- Events get pushed into the `Events` array
- `EventTypes` are stored for later usage
- TTL Index on the update date
- Schema version and validation results

Stats:

- 25 Mio. documents
- 150 GB in size
- 200 - 400 events per document
- Up to 20 msg/sec

Master Data Handling

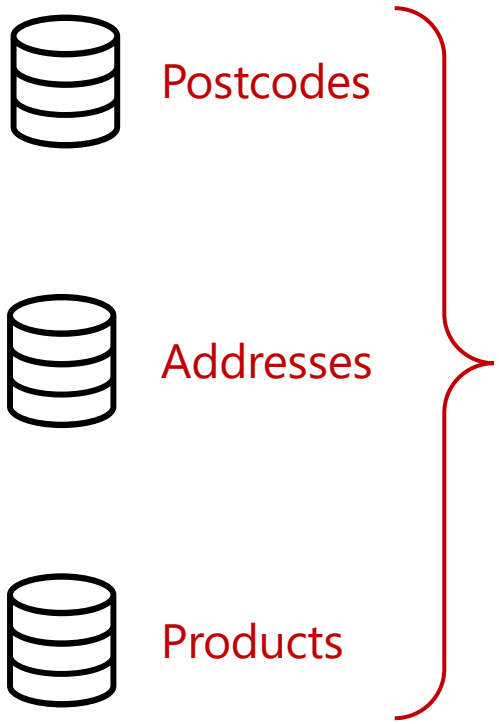
Master Data handling

Goals:
- Enable diagnostics with older master data
- Easy roll-back to a previous master data set
- Consume new master data set while the previous one is still in use

Challenges:
- Big datasets multiplied by number of releases to keep
- Lookups need to be quick, no matter how many releases we keep
- Cleanup of old release must not impact lookup performance

Solution:
- Partition master data by release version (prefixed with version)
- Leverage MongoDB dynamic collection creation
- Dropping collections to cleanup old releases

Master Data Handling



Master Data Release Bundling

Version 315

Version 314

MongoDB

Collections:

- 315-Postcodes
- 315-Adresses
- 315-Products

- 314-Postcodes
- 314-Adresses
- 314-Products

…

- 311-Postcodes
- 311-Adresses
- 311-Products

Cleanup:
Drop releases older than 312

Our journey to MongoDB Atlas

Our journey to MongoDB Atlas

- Need for flexibility because of nature of data
- Read-heavy workload
- Swiss wide rollout requires scalability
- 24/7 sorting requires zero downtime

- AWS DocumentDB was our natural choice
- Compatibility issues, missing features and resources
- Switch to MongoDB Atlas

DIE POST
FRAGILE
VORSICHT

Thank you
very much